

LESSON PLAN	COURSE: MDM4U
Date:	Unit of Study: Unit 1
Materials/Resources: Mathmetics of data management study guide	Lesson Focus/Theme: Chapter 5: Combination and the Binomial Theorem
Prior Knowledge:	
Overall Expectations: By the end of this lesson, students will: Solve problems involving the probability of an event or a combination of events for discrete sample spaces	
Specific Expectations: 1.4 determine, through investigation using class-generated data and technology-based simulation models (e.g., using a random-number generator on a spreadsheet or on a graphing calculator; using dynamic statistical software to simulate repeated trials in an experiment), the tendency of experimental probability to approach theoretical probability as the number of trials in an experiment increases (e.g., "If I simulate tossing two coins 1000 times using technology, the experimental probability that I calculate for getting two tails on the two tosses is likely to be closer to the theoretical probability of $\frac{1}{4}$ than if I simulate tossing the coins only 10 times") 1.5 recognize and describe an event as a set of outcomes and as a subset of a sample space, determine the complement of an event, determine whether two or more events are mutually exclusive or non-mutually exclusive (e.g., the events of getting an even number or getting an odd number of heads from tossing a coin 5 times are mutually exclusive), and solve related probability problems [e.g., calculate $P(\tilde{A})$, $P(A \text{ and } B)$, $P(A \text{ or } B)$] using a variety of strategies (e.g., Venn diagrams, lists, formulas) 1.6 determine whether two events are independent or dependent and whether one event is conditional on another event, and solve related probability problems [e.g., calculate $P(A \text{ and } B)$, $P(A \text{ or } B)$, $P(A \text{ given } B)$] using a variety of strategies (e.g., tree diagrams, lists, formulas)	

Learning Goals:

By the end of this lesson, students will be able to solve problem with combinations and also know the binomial theorem

Success Criteria:

By the end of this lesson, I will be able to solve problem with combination

By the end of this lesson, I will be able to use the binomial theorem

Assessment Strategies: Tests, quiz and project

Time:
10 min

Introduction/Minds-On:

Answer questions students don't understand from last class if any

Time:
100 min

Lesson:

Chapter 5: Combination and the binomial theorem:

5.3 Problem solving with combination

5.4 The Binomial theorem

Time:

10 min

Extension / Wrap-up:

Review the key concepts and formula for this lesson

Notes / Reminders / Homework:

Review for the test

Lesson Reflection: